

Computer Science Course — ENS Year 1 (EN)

****Instructor:**** BELACEL Madani

****Institution:**** Higher Normal School of Mostaganem

****Module:**** Computer Science

****Level:**** ENS — Year 1

****Duration:**** 1h30

****Academic year:**** 2025-2026

1) Learning outcomes

By the end of this module, ENS Year 1 students will be able to:

- Understand ****computer system architecture**** (CPU, memory, I/O, bus).
- Master ****operating system**** and file management basics.
- Design simple ****algorithms**** and analyze elementary complexity.
- Use fundamental ****data structures**** (arrays, lists, stacks).
- Understand ****networking**** principles (IP, DNS, HTTP) and basic ****security****.
- Get started with ****Python programming**** for teaching and research.

2) Course outline

1. Foundations and information representation
2. Machine architecture and execution cycle
3. Operating systems and processes
4. Algorithms and pseudocode
5. Basic data structures
6. Introduction to networks and the Internet
7. Cybersecurity and digital ethics
8. Python introduction and Year 1 project

3) Detailed content

Chapter 1 — Information representation ***(10 min)***

Binary, hexadecimal, ASCII/UTF-8 encoding. Signed integers (two's complement — overview). Images and sound (sampling intro).

Chapter 2 — Architecture ***(15 min)***

Von Neumann model: ALU, control unit, registers, memory, bus. Fetch-decode-execute cycle. Memory hierarchy (cache, RAM, disk).

Chapter 3 — Operating systems *(15 min)*

OS role: hardware abstraction, multiprogramming. Processes vs threads. Files, directory tree, permissions. Command line (cd, ls, mkdir).

Chapter 4 — Algorithms *(15 min)*

Variables, conditions, loops. Structured pseudocode. Complexity $O(n)$, $O(n^2)$ — intuitive intro. Execution tracing.

Chapter 5 — Data structures *(10 min)*

Array, linked list (diagram), stack (LIFO), queue (FIFO). Structure choice by problem.

Chapter 6 — Networks *(10 min)*

Client-server model. IP, DNS, URL. HTTP/HTTPS. Email, institutional cloud.

Chapter 7 — Security *(10 min)*

Confidentiality, integrity, availability. Passwords, MFA, phishing. 3-2-1 backup rule. GDPR and student data (awareness).

Chapter 8 — Python and project *(5 min)*

Variables, types, `if`/`for`, functions. IDLE or VS Code environment. **Project:** pedagogical Python script (quiz, grade average).

4) Assessment

Component	Weight
MCQ + paper algorithms	30 %
Python lab	35 %
Project + short presentation	35 %

5) Bibliography

- Cormen et al. — *Introduction to Algorithms* (introductory chapters).
- Tanenbaum — *Structured Computer Organization*.
- Python 3 documentation (docs.python.org).